Houdini: Security Benchmarking of Container Confinement

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***Abstract*—While container-based workloads are now a standard part of our cloud infrastructure, container security remains a challenging problem. Container confinement is a particularly pressing problem, as without it, a single vulnerable application can be used to compromise entire clusters of containers. While we have many technologies that can be used to secure containers, currently there is no easy way to determine whether a given Docker configuration provides even a basic level of protection. Here we present Houdini, a security test suite for container confinement. Much as network scanning tools can help catch misconfigured firewalls, Houdini can check whether a given Docker container configuration properly enforces confinement. While it can be used to test deployable containers, Houdini is optimised for testing and comparing container confinement technologies. Here we present the motivation, design, implementation, and initial results on running Houdini on a set of different container configurations. By providing a benchmark framework by which container confinement technologies can be evaluated, we believe Houdini can help foster the development of next-generation container confinement technologies.**

***Index Terms*—**

I. Introduction

Container-based workloads have become a cornerstone of modern cloud infrastructure, revolutionizing how applications are developed, deployed, and scaled, particularly within microservices architectures and DevOps workflows. Unlike hypervisor-based virtual machines (VMs), which each require a full operating system and rely on a hypervisor for hardware virtualization, containers are lightweight, share the host system's kernel, and incur minimal overhead. This efficiency makes them exceptionally well-suited for continuous integration and rapid scaling, allowing organizations to optimize resource utilization and reduce costs while ensuring quick, consistent deployments. Furthermore, containers enable seamless portability by packaging all userspace dependencies into a unified image, thereby avoiding the common problem of environment-specific discrepancies. Yet, despite containers' numerous benefits, they also introduce new security challenges, especially where container confinement is concerned.

Linux kernel features such as cgroups, namespaces, and mount points provide resource, process, and network isolation for containers. However, this separation is not foolproof. Kernel vulnerabilities, such as buffer overflows or privilege escalation flaws can allow a process to circumvent the very mechanisms designed to confine it. Misconfigurations can similarly open holes in container security, for instance by granting unnecessary capabilities, mounting sensitive host directories, or running containers in privileged mode. Finally, sophisticated attacks like container escapes exploit these weaknesses to break out of the container's resource and namespace isolation. Once an attacker gains elevated privileges or direct access to the host system, they can disrupt other containers, exfiltrate data, or compromise the entire infrastructure. Technologies like SELinux, AppArmor, and seccomp provide additional layers of security to avoid these issues, but are notoriously complex to configure and prone to misconfiguration. A best practices guide for container security [1] can help with configuring a container, but it offers no guarantee that the given configuration will perform as intended by the end-user. As container adoption skyrocketed across industries, particularly in multi-tenant cloud environments, these security challenges have become more pronounced.

To address the aforementioned challenges, the cloud industry has developed technologies like gVisor and Kata Containers [2], which isolate containers further by running them within lightweight VMs. While this approach is effective, gVisor-based containers bypass seccomp confinement by intercepting and emulating syscalls instead of directly blocking them through seccomp, which could result in security vulnerabilties if the emulation logic is flawed. By developing methods to audit and validate confinement mechanisms, container security could be simplified by reducing reliance on such solutions.

Test suites play a crucial role in verifying that systems meet specified requirements and function correctly under defined conditions. They are widely used across computing disciplines, from performance benchmarking to security assessment. For instance, the Standard Performance Evaluation Corporation (SPEC) [3] is best known for developing industry-standard benchmarks that evaluate CPU performance and power efficiency, while the Transaction Processing Performance Council (TPC) [4] establishes benchmarking standards for database performance. The Phoronix Test Suite [5] is a widely used benchmarking tool, particularly for Linux-based systems, providing automated performance evaluations across various hardware and software configurations.

Beyond performance testing, test suites are essential in software development to prevent regressions. Many organizations mandate that all code changes pass a comprehensive suite of tests before integration. In production environments, test suites help maintain reliability through uptime monitor-

ing, ensuring that services remain operational. Additionally, specialized security-focused test suites play a vital role in detecting vulnerabilities in deployed systems. Tools such as network scanners aid in identifying misconfigurations, outdated software versions, and unintended service exposures, helping organizations proactively mitigate security risks.

Despite the widespread use of test suites in the aforementioned areas, a significant gap remains in the context of containers. There are currently no standardized or widely adopted test suites designed specifically to evaluate container confinement, functionality, or isolation effectiveness. This absence leaves organizations to rely on ad hoc methods, which are often inadequate for uncovering subtle misconfigurations or ensuring robust confinement. Given the growing reliance on containerized environments in modern infrastructure, the development of comprehensive test suites tailored for containers is imperative. Such test suites would not only address these critical gaps but also provide a reliable foundation for improving trust and security in containerized systems.

In this paper, we describe Houdini, the first test suite for verifying container confinement. Given a Docker container configuration (including the kernel, docker, and runc version), Houdini will instantiate the container in a standalone QEMU-based VM and perform multiple tests (or tricks, as we call them) to evaluate whether the container is susceptible to various forms of misconfigurations or failures. These tests explore vulnerabilities that could disrupt container functionality, such as, but not limited to improper resource allocation, inadequate isolation, or insecure filesystem setups. Houdini is written in Python and is easily extensible, providing an extension language for writing controlled test cases (which we refer to as "tricks") designed to evaluate whether a container confinement mechanism properly enforces security boundaries.

In this paper, we describe the motivation, design, and implementation of Houdini. We also present the results of case studies by showing how Houdini can be used to detect misconfigured containers. Our hope with Houdini is that it will help with the deployment of better confined containers and will support the development of new technologies to more reliably confine containers.

The rest of this paper proceeds as follows. In Section III, we explain our motivation, which includes the container confinement problem. Section VIII describes related work. We describe Houdini's design in Section V. We describe the implementation of Houdini in Section VI. In Section VII we present case studies showing how Houdini can detect basic misconfigurations, and explain the tricks Houdini currently implements and their associated vulnerabilities. Section IX concludes our paper.

II. The Container Confinement Problem

Container confinement can be defined as the set of mechanisms and strategies used to isolate a containerized application from the host system and other containers. Effective confinement is critical for maintaining system security and stability, as it ensures that containers operate within defined boundaries by restricting their access to system resources, processes, networks, inter-process communication (IPC), filesystems, devices, and kernel interfaces.

A challenge with container confinement is that the level of confinement of a container is not fixed; it can vary based on the specific requirements or intentions of the user. In other words, the degree to which a container is isolated from the host system or other containers is customizable to meet the user's needs. At one end of the spectrum, you can lock a container down so tightly that it is almost entirely isolated from the host system, which minimizes potential attack surfaces but can also render the container incapable of performing key tasks. On the other hand, allowing a container too much freedom undermines the security benefits of containerization by exposing more privileges than necessary. The optimal configuration, however, is one that strikes a balance between maximum confinement (restricting access to system resources as much as possible) and function flexibility (enabling the container to perform its intended tasks). Achieving this balance is the core problem and goal of container confinement.

A standard methodology for assessing container confinement is crucial due to the complexity of modern systems and the intricate vulnerabilities they present. A formal approach involving rigorously evaluating and analyzing the security of the container system might seem ideal, however, it is impractical because container environments are highly complex, with interactions between kernels, runtimes, orchestration tools, and external dependencies that are difficult to model comprehensively. Moreover, many vulnerabilities stem from subtle details that a formal model might overlook. Instead, an empirical approach offers a practical alternative. Though it will never be exhaustive, the goal is not comprehensive coverage but rather establishing a baseline set of tests that demonstrate effective container confinement. This approach is achievable and can evolve over time as new vulnerabilities are identified, allowing the methodology to remain current. Starting with tests for isolation, privilege escalation prevention, and resource limitation enforcement provides a solid foundation. By focusing on demonstrating security effectiveness in critical areas, this methodology can enhance confidence in container confinement while enabling continuous improvement and adaptation to the evolving threat landscape.

III. Motivation

Houdini is designed to provide a systematic and reproducible approach to testing container confinement mechanisms. One of its primary goals is to evaluate whether container security features, such as namespaces, cgroups, seccomp, and Linux capabilities are effectively enforcing isolation. Many security tools rely on theoretical guarantees or static analysis, but Houdini focuses on empirical validation by executing controlled security tests that verify the correct enforcement of security mechanisms. By doing so, Houdini ensures that container security mechanisms are correctly configured.

One of the goals of Houdini is not only for testing container isolation but also as a means to optimize container configurations. By systematically simulating various tasks within different Docker containers, Houdini can help determine the minimal set of privileges required for a container to perform its intended function. For example, administrators can start with a container configured with broad privileges and then incrementally reduce permissions, such as Linux capabilities, seccomp filters, or resource limits, while monitoring task success. Houdini's testing framework will identify the point at which functionality is maintained while unnecessary privileges are removed, thereby pinpointing the least privilege configuration that still allows the container to operate effectively. This approach is highly beneficial because it adheres to the principle of least privilege. Minimizing the privileges granted to a container reduces its attack surface, making it significantly harder for an attacker to exploit vulnerabilities.

Although runtime monitoring and anomaly detection, such as with eBPF-based tracing can provide immediate insights into container behavior, our goal isn't to implement an exhaustive, continuous surveillance system. Instead, we aim to focus on auditing and validating the underlying isolation mechanisms to ensure they are correctly configured and consistently enforced.

The following are additional goals of Houdini:

Identifying Gaps Between Intended Confinement Policies and Actual Enforcement: There is often a discrepancy between security policies that administrators configure and what is actually enforced at runtime. For example, a security policy may specify that a container should not have network access, but due to a misconfigured Docker setting, the container may still be able to establish outbound connections. Similarly, a container may be configured with resource constraints (e.g., CPU and memory limits via cgroups), but in practice, those constraints may be ineffective or bypassable. Houdini is able to systematically test these security policies against real execution environments to uncover where enforcement fails. By detecting these discrepancies, organizations can adjust their configurations and improve security postures, ensuring that containers remain properly confined under expected conditions.

Challenging Misleading Security Assumptions with Empirical Testing: Container security is often discussed in terms of theoretical guarantees, with many security best practices based on assumptions rather than direct testing. For instance, documentation may state that a container running in unprivileged mode cannot escalate to root, but Houdini actively tests whether that is actually the case under different container configurations. By running real-world security validation instead of relying solely on security claims, Houdini helps challenge misleading assumptions and exposes where security mechanisms fail in practice. This is particularly important as container escapes and privilege escalation exploits continue to emerge, demonstrating that security controls do not always work as expected. By grounding security claims in measurable results rather than theory, Houdini helps researchers and practitioners develop more robust container security strategies.

In order to achieve the aforementioned goals, we designed Houdini with the following in mind:

1) *Reproducible Results.* A key goal of Houdini is to ensure that tests yield consistent and repeatable results. By controlling system variables and maintaining a structured testing environment, Houdini allows researchers and practitioners to compare results across different system configurations and security policies.
2) *Separation from the Host.* Houdini is designed to run tests inside a controlled containerized environment within a virtual machine (VM), ensuring that even if a trick triggers a security vulnerability, it does not compromise the host system running the tests. This design choice enhances the safety of security evaluations, preventing unintended side effects on the underlying infrastructure.
3) *Test Case Expressiveness.* Houdini tricks should be maximally expressive, such that some combination of steps can be used to achieve and test any desired result. It should be possible to define a new Houdini trick and modify existing tricks without modifying the Houdini binary. Moreover, it should be clear from reading a defined trick precisely what steps are involved, the consequences of each step passing or failing, and the overall nature of the exploit being tested.
4) *Focus on Observing Failures, Not Preventing Them.* Houdini is built to identify security weaknesses rather than defend against attacks.

IV. Security of the Testing Environment

If Houdini were compromised during testing, it would not necessarily pose a threat to the host system due to the design of the testing environment. Houdini is intentionally designed to operate within a controlled, containerized environment inside a VM. This means that even if a trick were to exploit a vulnerability within the containerized environment, the impact would be contained within the VM, isolated from the host system and other containers. The VM itself serves as an additional layer of security, providing a buffer that prevents the compromise from reaching the host infrastructure.

V. Design

Houdini is built using Python 3 and leverages a QEMU KVM VM to provide a controlled testing environment for container security validation. Instead of running containers directly on the host system, Houdini spins up a dedicated VM, ensuring that any security tests remain isolated and do not affect the host system.

At the core of Houdini's architecture, a Flask-based server runs on the host operating system. This server acts as a bridge between the host and the QEMU KVM VM, facilitating communication between them. The Flask server is responsible for sending commands and configuration data to the guest VM and receiving trick execution results. When a trick is initiated, the Houdini client on the host instructs the VM to start a new

trick. The VM then launches a container with the specified security configurations and executes the trick inside it.

Once the trick completes, the VM collects the results and sends them back to the host via the Flask server. The host then processes this data, logs the results, and determines whether the trick passed. A trick in Houdini is a structured security check designed to evaluate whether a containerized environment enforces confinement properly. Each test systematically probes specific security controls (e.g., namespaces, cgroups, capabilities) to determine if a container remains within its intended isolation boundaries or if a misconfiguration, vulnerability, or unexpected behavior allows for unintended behaviour such as, but not limited to privilege escalation, resource leakage, or escape.

This structured workflow ensures that Houdini remains independent of the host system's container runtime while providing a reliable and repeatable environment for testing container confinement mechanisms.

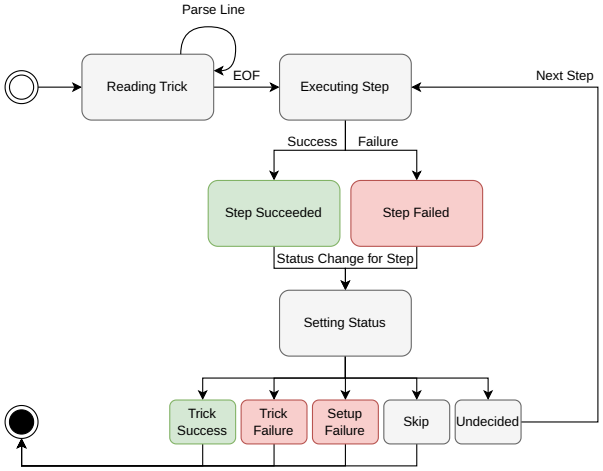


Fig. 1. A state machine diagram of Houdini running a Trick.

VI. Implementation

Tricks are divided into three separate files: a configuration file, a Python file, and a Dockerfile. Each `Houdini` trick begins with a configuration file (see Listing 1), which contains the Docker container's settings and environment configuration. The configuration file defines various Docker-specific parameters that are directly fed to the Docker API to start a container with the desired settings. Once the configuration is parsed, `Houdini` communicates these settings to the Docker API, using Docker's docker run command to initiate the container with these parameters. This ensures that the container is launched with the exact environment specified. This automation ensures the container is started exactly as specified, and that parameters such as network mode, mounts, security options, and user privileges are all controlled from a

Listing 1. Configuration file for a default Docker out-of-the-box security settings applied when a container is launched.

```
name: out-of-the-box docker configuration
dockerfile:
  - path: Dockerfile
dependencies:
  - server: False
docker_config:
  - network_mode: bridge
  - read_only: False
  - security_opt: []
  - pid_mode: null
  - cpu_shares: 1024
  - volumes : []
  - mem_limit: null
  - cpuset_cpus: null
  - cpu_quota: null
  - cpu_period: null
  - cap_add: []
  - cap_drop: []
  - privileged: False
  - user: root
  - pids_limit: null
  - ipc_mode: private
trick:
  - path: /houdini/tricks/HostMount
```

single file. This makes it easier to test, replicate, and share container environments.

The second component is a Python file that contains the trick implementation. This Python file is executed inside the container and is responsible for carrying out the trick scenario.

Listing 2. Exploit for CVE-2024-21616

```
import os

relative_path = '../../../../../'

try:
    os.chdir(relative_path)
    print(f"Successfully changed directory to:
↪ {os.getcwd()}")
except FileNotFoundError as e:
    print(f"Error: {e}")
except PermissionError as e:
    print(f"Error: {e}")
except Exception as e:
    print(f"Unexpected error: {e}")
```

Finally, the Dockerfile is used to configure and set up the container's environment. It specifies how the container should be built, including copying necessary files into the container and installing required dependencies. The Dockerfile also configures the container with the settings defined in the configuration file, ensuring the container is set up correctly to run the trick.

Listing 3.

```
FROM ubuntu:20.04
RUN apt-get update -y
WORKDIR /proc/self/fd/9
CMD ["bash", "-c", "ls
↪ ../../../../../houdini/tricks/HostMount"]
```

This modular design, comprising of the configuration file, Python script, and Dockerfile provides flexibility to define

and test various container escapes and vulnerabilities. It also allows users to easily customize or extend existing tricks by modifying any of the three components to suit their needs.

VII. Evaluation

To assess the effectiveness of container confinement mechanisms, we conducted a series of controlled experiments to evaluate whether common security policies successfully prevent resource starvation, unauthorized access attempts, and container escapes. Our evaluation focuses on testing various isolation features, including namespaces, cgroups, seccomp, and Linux capabilities, under different configurations and attack scenarios. These are done in the form of three case studies.

A. Case Study 1 - Evaluating Host Process Manipulation via PID Namespace Escape

In this evaluation, we investigate whether a Docker container running with `--pid=host` can interact with and manipulate host processes. The PID namespace is responsible for process isolation, ensuring that containerized processes remain separate from host processes. However, when `--pid=host` is enabled, the container shares the host's PID namespace, making all host processes visible inside the container. We assess the security implications of this setting by evaluating the following:

- Whether a container with `--pid=host` can list host processes.
- Whether the container can send signals `SIGKILL`, `SIGTERM` to host processes.
- Whether enabling `CAP_SYS_ADMIN` allows process tracing of host processes.
- Whether a container can read host process memory via `/proc/$PID/mem`.
- Whether a container with `--pid=host` can manipulate processes running inside another container that also has `--pid=host`.

The experiments were conducted using Docker running inside a QEMU VM to ensure an isolated and controlled test environment. The system specifications were as follows:

TABLE I
Testing Environment

Component	Specification
Host Machine	20 Core CPU, 16GB RAM, Ubuntu 24.04
Virtual Machine	QEMU, 4 Core CPU, 12GB RAM, No Distribution
Docker Version	27.0.3
Kernel Version	Linux 6.1
Container Runtime	`runc` 1.1.11

The container was started with various security configurations:

- Default Docker container configuration (separate PID namespace)
- Docker container with `--pid=host`
- Docker container with `--pid=host` and additional capabilities `CAP_KILL`, `CAP_SYS_PTRACE`
- Privileged Docker container with `--pid=host`
- Two separate containers, both with `--pid=host`, to evaluate cross-container process manipulation.

For each security configuration, different methods were tested to determine whether a process inside the container could interact with host processes or another container's processes.

The evaluation began by assessing whether a container with `--pid=host` could list host processes using `ps aux`. As expected, the container displayed all host processes, confirming that sharing the host PID namespace fully exposes the host process table.

Following this, we tested whether the container could send termination signals (`SIGKILL`, `SIGTERM`) to host processes using `kill -9 <PID>`. Without the `CAP_KILL` capability, the operation was not permitted. However, enabling `CAP_KILL` process termination within the namespace's limits, and granting `--privileged` access enabled the container to kill any host process.

Next, we investigated whether the container could attach a debugger to host processes via `strace -p <PID>`. Without `CAP_SYS_PTRACE`, the operation was denied, but enabling this capability successful attachment to a host process, raising concerns about potential leakage of sensitive data through system call tracing.

We then examined whether the container could read host process memory by attempting to access `/proc/<host_pid>/mem`. With Docker's default restrictions, the operation was denied, but when running the container with `--privileged`, the process memory could be read using `cat /proc/<host_pid>/mem`.

Finally, we tested cross-container process manipulation by launching two containers with `--pid=host`. A process running in Container A (e.g., `sleep 1000`) was visible to Container B, which was able to list and terminate it, proving that containers sharing the host PID namespace are not isolated from each other and can manipulate each other's processes.

The result was that the process inside Container A was visible to Container B, and it could be terminated using `kill -9 <PID>`, proving that containers with `--pid=host` share the same namespace and are not isolated from each other.

The effectiveness of different configurations in preventing host and cross-container process manipulation is also illutrated in Table II.

This case study illustrate an example scenario where the security policies administrators intend to implement are not properly enforced when the container is running. For example, an administrator may want to use the host's process namespace for tracing host processes but restrict the container from manipulating those processes. To achieve this, the administrator configures the container to share the PID namespace ('–pid=host'), allowing the container to trace processes but preventing any other interactions. Houdini validates this configuration by performing controlled tests to ensure the container can only trace processes and cannot send signals, attach debuggers, or read process memory. If Houdini detects

TABLE II
EFFECTIVENESS OF DOCKER SECURITY MECHANISMS AGAINST HOST PROCESS MANIPULATION

Security Configuration	List Host Processes	Send Signals	Attach Debugger	Read Memory	Cross-Container Manipulation
Default Docker (PID namespace isolation)	✗	✗	✗	✗	✗
`--pid=host` (no extra caps)	✓	✗	✗	✗	✓
`--pid=host` + `CAP_KILL`	✓	✓	✗	✗	✓
`--pid=host` + `CAP_SYS_PTRACE`	✓	✗	✓	✗	✓
Privileged Container (`--pid=host` + `--privileged`)	✓	✓	✓	✓	✓

that the container can perform any of these unauthorized actions, it reveals a gap between the intended policy and its actual enforcement. Houdini's tests provide actionable insights that help administrators adjust configurations to enforce proper isolation while allowing the container to trace processes. By systematically testing various configurations, Houdini helps administrators identify the minimal privileges necessary for the container to perform its intended tasks without compromising security.

B. Case Study 2 - Evaluating Docker Privileged Port Forwarding Restrictions

In this case study, we evaluate Docker's network isolation mechanisms to determine how well they enforce privileged port binding restrictions. Starting with Docker version 20.03, a significant change was introduced to Docker. They redefined unprivileged ports to start at 0 instead of 1024, effectively removing the need for root to bind to privileged ports. Table III presents the results.

This case study highlights how discrepancies between Docker versions in handling privileged port binding can create gaps between administrators' intended confinement policies and actual enforcement. A change in Docker's handling of binding to privileged ports, such as the removal of the NET_BIND_SERVICE requirement starting from Docker 20.03, may allow containers to bypass restrictions even when policies are set to prevent it.

For example, administrators might configure a container to be restricted from binding to privileged ports, but if Docker is running a version where the NET_BIND_SERVICE capability is no longer required for root containers (like in Docker 20.03 and beyond), the container could still bind to these ports despite the policy specifying otherwise. This creates a mismatch between the intended policy (restricting privileged port binding) and the actual enforcement (where containers can still bind to those ports in certain Docker versions).

Houdini helps identify these gaps by systematically testing container configurations across different Docker versions. By simulating different Docker versions and security policies, Houdini verifies whether containers are correctly enforcing the intended restrictions on privileged port binding. For instance, Houdini could test whether containers configured to prevent binding to privileged ports are still able to bind to those ports in Docker versions 20.03 or later, where the security model has changed. If a container running with Docker 20.03 or newer can bind to privileged ports despite the configuration specifying otherwise, Houdini will detect this discrepancy and highlight the gap between the intended policy and actual enforcement.

C. Case Study 3: Evaluating CVE-2024-21616

CVE-2024-21616 is a critical vulnerability affecting Docker and runc that allows malicious containers to escape their isolation and execute arbitrary commands on the host system. This poses a significant risk to enterprises relying on containerization, as attackers could gain unauthorized access to sensitive data and system resources.

The root cause of this vulnerability is the improper handling of a container's working directory when set to a special file descriptor path such as `/proc/self/fd/<fd>`, where `<fd>` typically points to `/sys/fs/cgroup`. If exploited, this flaw allows a containerized process to break out of its confined environment, effectively compromising the host machine. The vulnerability affects runc versions from `v1.0.0-rc93` to `1.1.11`.

To test the exploitability of CVE-2024-21616, we designed a Houdini trick that executes the following steps:

1) Launch a container with a vulnerable version of runc.
2) Set the working directory to `/proc/self/fd/<fd>`, pointing to a controlled file descriptor.
3) Attempt to break out of the container by leveraging the misconfigured working directory.
4) Validate whether commands can be executed on the host system.

As this vulnerability is due to the runtime itself and not related to the configuration of a particular container, We conducted tests across only various runc versions. This is shown in Table VII-C.

TABLE IV
RUNC VERSIONS AGAINST CVE-2024-21616

Runc Version	Exploit Success
runc 1.1.7	✓
runc 1.1.8	✓
runc 1.1.9	✓
runc 1.1.10	✓
runc 1.1.11	✓
runc 1.1.12	✗
runc 1.1.13	✗
runc 1.1.14	✗

VIII. RELATED WORK

At the time of this writing, to our knowledge, Houdini is the first comprehensive approach for evaluating and comparing *container confinement mechanisms*. Much of the container

TABLE III
EFFECTIVENESS OF DOCKER SECURITY MECHANISMS IN PRIVILEGE PORT BINDING

Docker Version	NET_BIND_SERVICE + root	NET_BIND_SERVICE + root-less	root	root-less	Default Container (Bridge Mode)
19.03.9	✓	✗	✗	✗	✗
19.03.0	✓	✗	✗	✗	✗
20.10.0	✓	✗	✗	✗	✗
20.10.3	✓	✗	✗	✗	✗
23.0.0	✓	✓	✓	✗	✗
25.0.0	✓	✓	✓	✗	✗
27.0.1	✓	✓	✓	✗	✗

security literature to date has focused on either vulnerability scanning of container images, building offensive/attack tools for container escapes, or proposing best practices for securing container runtimes. While these papers, tools, and documents do not directly share our research objectives, they still broadly fit into the container security landscape. Additionally, research on VM escapes and sandbox escapes has provided valuable insights into isolation vulnerabilities, many of which relate to container security. Techniques used to break out of hypervisors and sandboxed environments often share similarities with container escape methods, which the importance of strong confinement mechanisms. The remainder of this section does not aim to exhaustively enumerate all container security tools and systems, but rather to highlight the various salient methods, including prior research on times the confinement of VMs and sandboxes have been escaped as it is related to work.

Vulnerability Scanners. Many free, open, and closed-source tools exist for identifying the presence of known vulnerabilities in container images. Shu *et al.* [6] developed a vulnerability scanning tool to scan Docker Hub container images at scale. They found that on average, official and community images have concerning amounts of vulnerabilities (180+), and that these vulnerabilities remain unpatched for hundreds of days. Their study utilized the Docker Image Vulnerability Analysis (DIVA) framework, which automates the discovery, download, and analysis of over 300,000 Docker images. The findings revealed that more than 80% of both official and community images contained at least one high-severity vulnerability, and many images were not updated for extended periods. Moreover, vulnerabilities were often inherited from parent images to child images, further increasing security risks.

Best Practices. Users in search of (often high-level) advice on how to improve the security of their container deployments can consult one of many best practices guides available online. One such guide is the Center for Interent Security (CIS) Docker Benchmark [1]. In our experience, guides tend to offer generic advice (e.g., "*Only allow read access to the root filesystem*". This is problematic because it oversimplifies container security. Generic advice like "Only allow read access to the root filesystem" may seem straightforward, but it doesn't account for the specific context or unique configuration of an individual deployment. Such one-size-fits-all recommendations can lead users to believe that simply following these broad guidelines is enough to secure their systems, potentially leaving critical vulnerabilities unaddressed. In reality, effective container security requires a nuanced, tailored approach that considers the particular needs and threat models of each environment. Houdini fills the gap between generic advice and practical security by enabling you to objectively evaluate the effectiveness of your container confinement measures.

Offensive Tools. The offensive security community has developed a range of specialized tools aimed at identifying and exploiting vulnerabilities within containerized environments. Containers, while providing strong isolation, are not immune to attacks, and these tools focus on testing the robustness of container defenses. CDK[1] and DEEPCE[2] are two widely used, open-source container penetration testing toolkits that leverage a collection of known exploits to gain persistence, escape the container, and gather sensitive information about the container environment. These tools aim to bypass isolation mechanisms like namespaces, cgroups, and SELinux, often using a variety of aggressive techniques to identify weaknesses in container security. CDK specifically uses any available method to circumvent security measures, providing a broad toolkit to test the limits of a container's defense. Meanwhile, DEEPCE, developed by stealthcopter, focuses on privilege escalation and container enumeration, attempting to exploit specific weaknesses to achieve container escapes, thereby compromising the host system. In contrast, Houdini takes a more systematic and structured approach to container security testing. Rather than focusing solely on exploiting vulnerabilities, Houdini allows for the direct comparison of defensive techniques by clearly defining the testing environment, outlining the exploit steps, and documenting the expected outputs.

Virtual Machine Escapes VM escapes occur when an attacker successfully breaks out of a virtualized confined environment, gaining unauthorized access to the host system. Given that hypervisors manage multiple VMs on shared hardware, such escapes can lead to host compromise, cross-VM attacks, and lateral movement within cloud environments. One of the most notable VM escape vulnerabilities is VENOM (CVE-2015-3456) [7], which exploited a flaw in the virtual floppy disk controller (FDC) implementation in QEMU, affecting platforms like Xen, KVM, and VirtualBox. Other significant VM escape vulnerabilities include Cloudburst (2009), which exploited VMware's virtual graphics driver; CVE-2018-6981 and CVE-2018-6982, which targeted VMware ESXi, Workstation, and Fusion [8]; and multiple VirtualBox vulnerabilities (2017) enabling guest-to-host code execution [9]. Additionally, side-

[1]https://github.com/cdk-team/CDK
[2]https://github.com/stealthcopter/deepce

channel attacks like MDS/ZombieLoad (2019) demonstrated how speculative execution flaws could be leveraged across VMs [10]. More recent research has uncovered weaknesses in encrypted virtualization, such as the SEVered attack (2018), which extracted plaintext memory from AMD SEV-protected VMs [11], and the Heckler attack (2024), which manipulated non-timer interrupts to compromise confidential VMs in AMD SEV-SNP and Intel TDX [12].

Sandbox Escapes Sandboxing is a critical security mechanism for restricting the execution of untrusted code, commonly used in browsers, mobile apps, and security tools. A sandbox escape occurs when an attacker bypasses these restrictions, executing arbitrary code with higher privileges and potentially leading to full system compromise. Modern web browsers heavily rely on sandboxing to isolate JavaScript execution from the underlying OS. However, several high-profile vulnerabilities have enabled sandbox escapes. For example, CVE-2019-5786 [13], a use-after-free vulnerability in the FileReader API of Google Chrome, allowed attackers to escape Chrome's sandbox and execute code on the host. Other notable sandbox escape vulnerabilities include CVE-2022-26706, which exploited a flaw in macOS's App Sandbox to execute code outside of restricted permissions [14]; CVE-2023-36719, a stack corruption vulnerability in Windows OS that enabled escape from the Chromium sandbox [15]; and CVE-2024-4040, which was actively exploited to break out of CrushFTP's Virtual File System (VFS) [16]. Additionally, vulnerabilities in Judge0, an online code execution system, exposed systems to full takeover by allowing attackers to escape its sandbox [17]. Research has also uncovered critical flaws in vm2, a JavaScript sandbox library, which could be exploited to bypass its security restrictions and gain remote code execution [18].

IX. Conclusion

In this paper, we presented Houdini, a security benchmarking framework designed to evaluate the effectiveness of container confinement mechanisms. As container-based workloads continue to dominate cloud infrastructures, ensuring that containers are properly isolated is paramount for maintaining security. Houdini offers a systematic and empirical approach to test and assess various container security configurations, highlighting potential misconfigurations that may compromise container isolation. Our evaluation demonstrated the ability of Houdini to detect critical misconfigurations, including those leading to privilege escalation. By providing a reliable and reproducible method for testing container confinement, Houdini helps bridge the gap between theoretical security assumptions and practical real-world behavior.

As container technologies evolve, it is essential to continuously assess and improve their security mechanisms. Houdini's modular design ensures that it remains adaptable to new security challenges and configurations, allowing for ongoing validation and refinement of container security practices. By offering an open framework for container confinement benchmarking, Houdini aims to support the development of more secure container technologies, ultimately contributing to a safer cloud infrastructure for the future.

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